

Reactivity Differences of Rieke Zinc Arise Primarily from Salts in the Supernatant, Not in the Solids

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ABSTRACT: Contrary to prevailing thought, the salt content of supernatants is found to dictate reactivity differences of different preparation methods of Rieke zinc toward oxidative addition of organohalides. This conclusion is established through combined single-particle microscopy and ensemble spectroscopy experiments, coupled with careful removal or keeping of the supernatants during Rieke zinc preparations. Fluorescence microscopy experiments with single-Rieke-zinc-particle resolution determined the microscale surface reactivity of the Rieke zinc in the absence of supernatants, thus pinpointing its inherent reactivity independent of the convoluting supernatant composition. In parallel experiments, scanning electron microscopy, energy-dispersive spectroscopy, X-ray photoelectron spectroscopy, and inductively coupled plasma-mass spectrometry characterized the zinc metal chemical composition at the bulk and single-particle levels. Proton nuclear magnetic resonance spectroscopy kinetics characterized bench-scale Rieke zinc reactivity in the presence and absence of different supernatants and exogenous salt additives. Together, these experiments show that the differences in reactivity from sodium-reduced vs lithium-reduced Rieke zinc arise from the residual salts in the supernatant rather than the differing salt compositions of the solids. This supernatant salt also determines the structure of the ultimate organozinc product, generating either the diorganozinc or monoorganozinc halide complex. That different organozinc complexes formed upon direct insertion to different preparations of Rieke zinc was not previously reported, despite Rieke zinc's widespread use. These findings impact organozinc-reagent and nanomaterial synthesis by showing that, unexpectedly, desired Rieke zinc reactivity can be achieved through simple addition of soluble salts to solutions that were used to prepare the metals—a substantially easier synthetic manipulation than solid composition and morphology control.



INTRODUCTION

Organozinc compounds are workhorse reagents of modern synthesis, enabling C–C bond formation with high functional group tolerance.¹ Though they are considered classical reagents, their importance in chemistry continues to grow, with 1820 citations on the topic of “organozinc” in 2000 rising to over 4830 citations in 2021.^{2,3} As a route to generating organozinc reagents, freshly prepared zinc metal powder, made by in situ reduction of zinc salts by alkali metals (i.e., “Rieke zinc”), is highly reactive. It therefore provides access to organozinc reagents by direct insertion into carbon–halide bonds that are otherwise sluggish to react with or fully inaccessible from typical commercial zinc metal powder.⁴ Rieke zinc therefore plays a broad role in modern synthesis, from materials synthesis (e.g., Ti nanoparticles⁵) to drug discovery (e.g., via generation of organozinc reagents for use in Nobel-Prize recognized Negishi cross-coupling reactions^{6,7}). Since the early 1970s, it has been qualitatively known that the preparation method for Rieke zinc greatly impacts the reactivity of the resulting zinc powder, leading to a proliferation of different preparation methods (Figure 1).^{8–10} The mechanistic causes of these reactivity differences remain,

however, poorly understood.¹¹ Without the ability to understand the cause of variation in different batches of Rieke zinc,^{12,13} the rational selection of the best preparation by users is not straightforward, and the expansion of this useful chemistry to access new carbon–(pseudo)halide bond insertion, polymers,¹⁴ and nanomaterials is impeded.

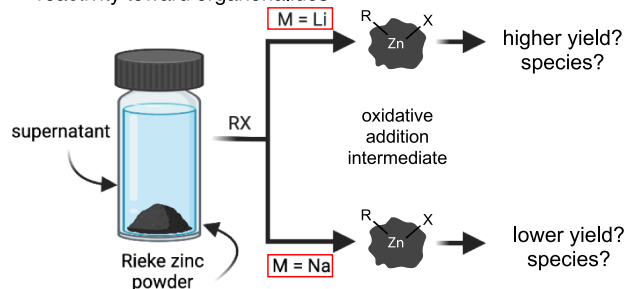
Rieke proposed several interwoven hypotheses for these differences in reactivity, most notably, that alkali metal ions, halogens, and other byproducts from reduction became entrapped in the metal lattice and accounted for the higher reactivity.¹⁵ Central to these hypotheses is the idea that the composition and physical features of the zinc solids are the cause of the differences in observed reactivity. This highly influential “solids” thought process has guided the field of

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a. Highly reactive Rieke zinc preparations produce differences in reactivity toward organohalides



b. Rieke's descriptions of differences in Rieke metal reactivity

Source of Rieke metal	Residual salt	Reactivity in organozinc reagent formation
ZnCl ₂ + 2 Li	LiCl	"Highly reactive" ⁹
ZnCl ₂ + 2 Li-naphthalenide	LiCl	"Even more reactive" than ZnCl ₂ + 2 Li ¹⁷
ZnCl ₂ + 2 Na	NaCl	Lower than ZnCl ₂ + 2 Li ⁹
ZnCl ₂ + 2 K	KCl	Lower than ZnCl ₂ + 2 Li ⁹
ZnBr ₂ + 2 Li	LiBr	Lower than ZnCl ₂ + 2 Li ⁹
Zn(CN) ₂ + 2 Li	LiCN	"More reactive" than ZnCl ₂ + 2 Li ¹⁰

Figure 1. (a) Schematic of the synthesis of Rieke zinc and plausible corresponding differences arising from reduction with lithium or sodium. (b) Table of Rieke's descriptions of the reactivity of Rieke zinc made with various reductants and Zn counterions. Preparations examined here shown in a red box.

development of active metals—both zinc and other metals—since Rieke's initial publications.^{4,8,9,16}

To add to the original reactivity mysteries, unfortunately, Rieke published primarily qualitative reactivity comparisons of different preparations. A portion of these assessments were based on reactivity with different substrates, for example, the ability of Rieke zinc formed by a given preparation method to undergo direct insertion with less reactive organochlorides vs the typically more reactive organoiodides or organobromides.^{4,8–10,15,17–24} While rational, such comparisons leave a gap in quantitative understanding (which is now partially addressed in this manuscript). We have compiled a collection of Rieke's reactivity assessments in Figure 1b.^{9,10,17} These reactivity assessments, though qualitative, currently appear to influence choices of other researchers. The reduction of ZnCl₂ via Li naphthalenide in THF solvent appears in our survey to be the most represented Rieke zinc synthesis,^{25,26} consistent with Rieke's assessment of it being "even more reactive."

Because of the central role of Rieke zinc in synthesis, the mechanism(s) behind these differences in reactivity has been subject to much speculation but so far all centered around differences in the solids themselves.^{16,27} Indeed, the solids are physically and compositionally different depending on the preparation, so this original line of thinking is rational. Prior bulk analysis established that the Rieke zinc particles are complicated mixtures that contain embedded salt byproducts from the reduction step; thus, Rieke postulated that differences between the high surface area and the different residual salts embedded in the crystal lattice of zinc from the byproducts of reduction differentially assisted in rapid oxidative addition.¹⁵ However, we note that the presence of simple differences in the solid composition or morphology of the solid is not

necessarily the cause of reactivity differences (or even if they are a cause, that they are the sole or predominant cause).

In previous studies from our laboratory, fluorescence microscopy with single-zinc-particle resolution demonstrated the mechanism by which DMSO²⁸ and exogenous LiCl^{29–32} accelerate the rate of formation of organozinc compounds from commercial zinc metal powders and organohalides. This formation proceeds through a two-step mechanism: (1) surface oxidative addition followed by (2) solubilization of the resulting organozinc surface intermediate. Prior to the establishment of single-molecule and single-zinc-particle fluorescence microscopy as a tool to study organic reactions, this two-step mechanism was unknown.³³ It was challenging to identify and study due to the inability to detect small quantities of oxidative addition intermediate building up on the zinc surface, which did not build up to quantities sufficient for detection by traditional ensemble analytical techniques. With subensemble fluorescence microscopy, however, it became possible to image and compare the behavior of small quantities of surface intermediates between different sets of conditions.³⁰ That is, it became possible to perform traditional physical-organic and mechanistic studies³⁴ on this previously unknown intermediate. Using this approach, we previously discovered that this surface oxidative addition intermediate, when formed by reaction between an alkyl iodide and a commercial large-particle-size zinc metal surface, was persistent in the absence of LiCl and therefore observable on the surface of zinc metal via fluorescence microscopy, down to the single-zinc-complex level. Addition of LiCl to the intermediate resulted in solubilization and release from the metal surface, the second step of the mechanism.^{29–31}

Given the mechanistic role identified for solution-phase LiCl in accelerating the reaction rate of organohalides with these commercial zinc powders,³² we wondered if solution-phase LiCl may play a previously overlooked and defining role in the reactivity of Rieke zinc preparations. Because LiCl is a byproduct of the Rieke zinc synthesis when lithium metal is the reductant with ZnCl₂ (Figure 1b, entry 1), Rieke had attributed the enhanced reactivity of this preparation to salt impurities within the solid crystal lattice.¹⁵

We now considered that the sensitive fluorescence imaging technique may assist in identifying the mechanistic source of the reactivity of different preparations of Rieke zinc, an unexplored area for fluorescence imaging. In particular, what accounts for the reactivity differences between entries 1 and 2 in Figure 1b (the lithium-reduced and sodium-reduced Rieke zinc)? Are the reactivity differences caused by the solids or by the supernatants?

We here develop and apply a combined approach of highly sensitive single-Rieke-zinc-particle fluorescence microscopy imaging with scanning electron microscopy (SEM), energy-dispersive spectroscopy (EDS), inductively coupled plasma-mass spectrometry (ICP-MS), and proton nuclear magnetic resonance (¹H NMR) spectroscopy. This combined analytical approach produces a unique combination of microscale and macroscale information that now enables correlation of solid and solution composition with reactivity. This correlation was not available through prior analytical techniques. Careful studies separate supernatant and metal powders for individual analysis. Thus, the roles of the solids and the roles of the supernatants are separated and understood at a granular and previously unobtainable mechanistic level. The solids are examined for their inherent ability to undergo oxidative

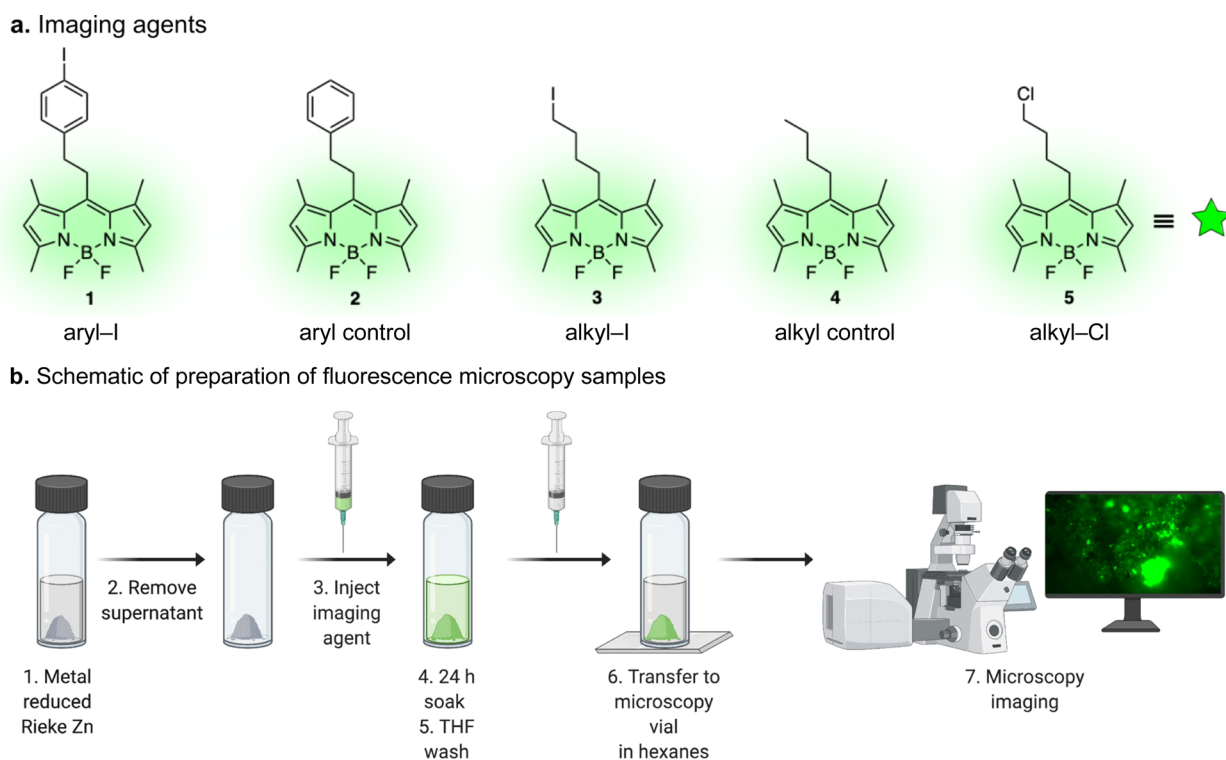


Figure 2. (a) BODIPY-based imaging agents used in Rieke zinc experiments. (b) Schematic of procedure for imaging.

addition with alkyl iodides by fluorescence microscopy, away from their respective supernatants. The resulting unique union of information enables characterization of otherwise unobservable organozinc intermediates on the surface of Rieke zinc from different preparations for the first time along with ^1H NMR kinetics of bulk reaction rates in the presence and absence of supernatants.

Contrary to prevailing thought, the composition of the supernatant plays the dominant role in reactivity differences between lithium-reduced and sodium-reduced Rieke zinc. This finding is exciting because solution compositions are far easier to control (by simple addition of soluble additives)³⁵ than solid lattice compositions—especially lattice compositions of solids that arise from ill-defined, highly heterogeneous reaction mixtures such as those occurring in Rieke metal preparation.

Furthermore, these data show that the ultimate solution-phase organozinc reagents made from differently prepared Rieke zinc are structurally different, which to our knowledge was not previously reported. Thus, the organozinc products from different procedures likely have different downstream reactivity, in line with previous salt studies by Organ.³⁶

RESULTS AND DISCUSSION

Selection of Imaging Agents. The boron dipyrromethene (BODIPY) fluorophores shown in Figure 2a (1, 3, 4) were synthesized according to literature procedures and have been shown in previous studies to contain chemically inert “spectator” fluorophore probes that are suitable for imaging the oxidative addition process and intermediate on zinc surfaces.^{30,31} Oxidative addition reactivity occurs through the distal carbon–iodide bond. Imaging agents 2 and 5 have been designed and synthesized new for this work (see the SI). This suite of imaging agents 1–5 include an oxidative-addition reactive sp^2 /aryl iodide 1, with an otherwise similar control

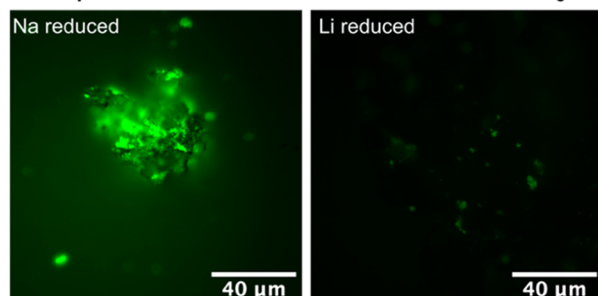
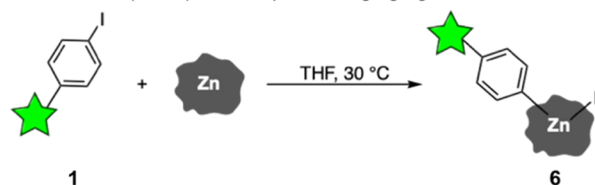
imaging agent 2 that lacks the iodide and therefore cannot undergo oxidative addition to the zinc surface; oxidative-addition reactive sp^3 /alkyl iodide 3, with its otherwise similar control imaging agent 4 that lacks the iodide and therefore cannot undergo oxidative addition to the zinc surface and challenging oxidative-addition partner sp^3 /alkyl chloride 5, with previous control 3. On the basis of reported reactions of Rieke zinc with organohalides, we anticipated that the order of oxidative addition reactivity of the organohalides with the zinc surface is alkyl iodide⁹ > aryl iodide³² > alkyl chloride (which to our knowledge appears to require a second metal catalyst or a synthesis that makes cyanide as a byproduct).¹⁰

Fluorescence Microscopy Studies of Reactivity of Isolated Rieke Zinc Solids. A series of highly sensitive fluorescence microscopy studies with single-zinc-particle resolution were performed to investigate which preparations of solid Rieke zinc resulted in the formation of persistent surface oxidative addition products, with which imaging agents, and at which temperatures. We previously showed that this analytical technique has sufficient sensitivity to image single organometallic complexes on zinc surfaces,³¹ providing confidence that, if oxidative addition intermediates were present, the current technique would be suitable for detecting them.

Fluorescence microscopy samples for imaging were prepared under an inert argon or nitrogen atmosphere in a glovebox (Figure 2b). Rieke zinc was prepared according to literature preparations,¹⁵ and the solids were separated from the supernatant by allowing the solution to settle overnight. To limit variables, both sodium and lithium reductions were performed similarly using naphthalene. Then, the supernatant, which contained all soluble byproducts, was removed, and the residual solids were washed with clean THF. These were the key steps for separation of the solids from residual salts in the supernatant.

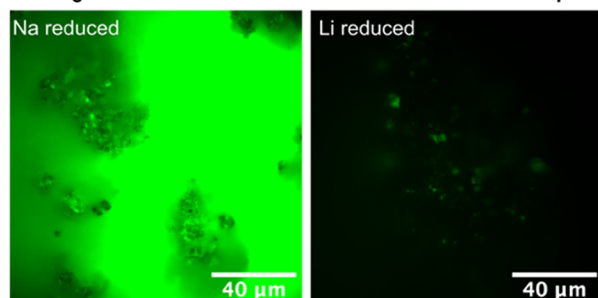
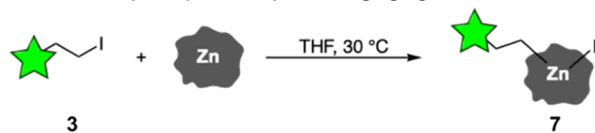
Oxidative addition intermediate image on zinc reduced by Na

a. Rieke zinc (no supernatant) with imaging agent 1



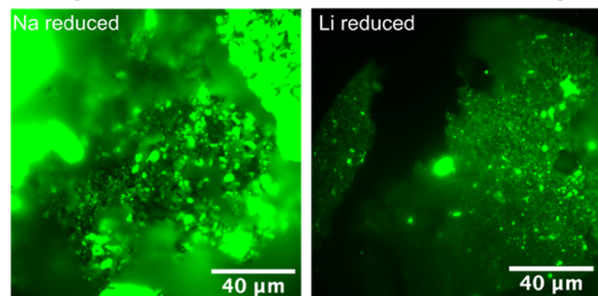
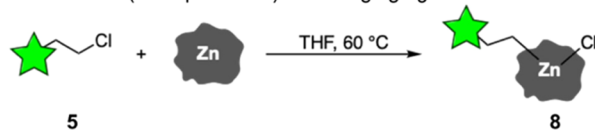
6 observed 6 not observed

c. Rieke zinc (no supernatant) with imaging agent 3



7 observed 7 not observed

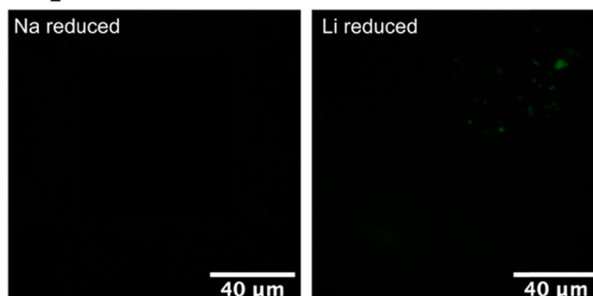
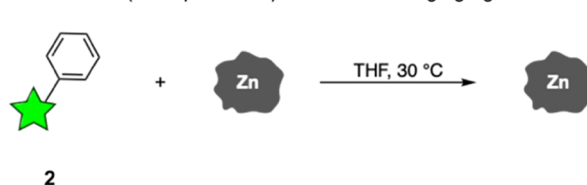
e. Rieke zinc (no supernatant) with imaging agent 5



8 observed 8 not observed
assigned as physisorption
see Figure 2g

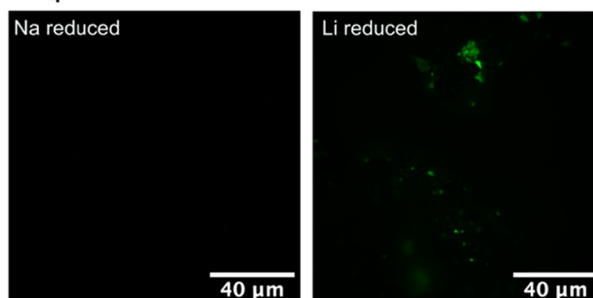
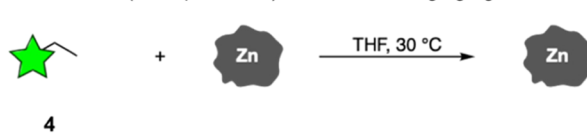
Control reactions

b. Rieke zinc (no supernatant) control with imaging agent 2



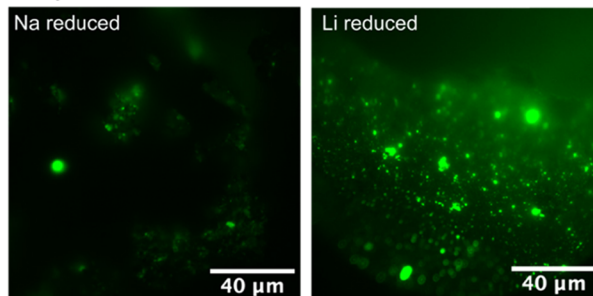
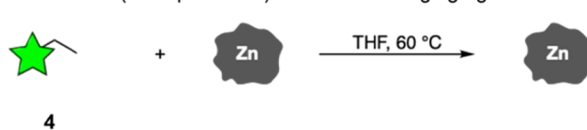
no physisorption no physisorption

d. Rieke zinc (no supernatant) control with imaging agent 4



no physisorption some physisorption

f. Rieke zinc (no supernatant) control with imaging agent 4



some physisorption
observed physisorption observed

Figure 3. (a–f) Fluorescence microscopy images of lithium- vs sodium-reduced Rieke zinc and the corresponding chemical equations, compared with corresponding control experiments. Images (a–d) are acquired with identical camera settings and laser power and are displayed at the same brightness–contrast settings to allow for direct comparison of (a–d). Samples in (e) and (f) were acquired at a higher laser power than (a–d), but they were acquired with the same camera, and displayed at the same brightness–contrast settings as (a–d). This allows direct comparison between (e) and (f). Samples were imaged in hexanes. All samples did not include the supernatant from their respective preparation methods. Replicate data available in the SI.

These microscopy experiments were prepared by weighing solid Rieke zinc in a vial and adding an imaging agent solution of **1**, **2**, **3**, **4**, or **5**. (Figure 2b). The mixture in the vial was swirled and allowed to react at the designated temperature for 24 h. After 24 h, the imaging agent solution was removed by a plastic syringe, and the Rieke zinc powder was quickly washed 3× with clean THF to remove residual imaging agent solution. This step was necessary to produce a dark background solution suitable for imaging. Finally, the Rieke zinc powder was transferred to a gas-tight microscopy imaging vial in hexanes, removed from the glovebox, and imaged on an inverted microscope in widefield epifluorescence mode (Figure 2b). Hexanes were used for the final imaging step because they preserved the status of oxidative addition intermediates on the surface of the Rieke zinc; if hexanes were not used, intermediates were found to be partially soluble in THF and to release from the surface of Rieke zinc upon prolonged soaking in THF.

To assess the validity of experiments using oxidative-addition reactive compounds **1**, **3**, or **5**, otherwise identical experiments with control imaging agent **2** or **4** were performed. The control imaging agents were used to assess whether the imaging agent was merely physisorbed onto the zinc surface, without formation of a chemical bond through oxidative addition into the carbon–iodide bond, because the control imaging agents lacked carbon–iodide bonds. If bright green fluorescent “hotspots” were formed with the organohalide imaging agent (**1**, **3**, or **5**) but not with the corresponding control imaging agent (**2** or **4**), then the green “hotspots” were assigned as oxidative addition intermediates.^{28–31} However, if the corresponding control experiment displayed similar green “hotspots”, then it was concluded that the surface of the zinc physisorbed both the organohalide imaging agent and the control imaging agent similarly and that there is no evidence to support oxidative addition intermediates (i.e., the signal is consistent with physisorption). Control and noncontrol images were acquired with identical camera settings and laser power and displayed at identical brightness–contrast settings to enable direct comparisons between samples. In all cases, images show representative data from a survey of the sample. All imaging experiments were performed in replicates, and replicates show the same trends. Replicate data are available in the SI.

The series of solid reactivity is shown in Figure 3. The first set of experiments, in Figure 3a, compares sodium-reduced Rieke zinc (left) and lithium-reduced Rieke zinc (right), both reacting with aryl iodide oxidative-addition partner **1**. Sodium-reduced Rieke zinc (Figure 3a, left image) displays a high concentration of fluorescent bright green “hot spots,” and lithium-reduced Rieke zinc (Figure 3a, right image) is dark with a small number of fluorescent “hot spots.”

The corresponding set of control reactions with imaging agent **2** are shown in Figure 3b. Sodium-reduced Rieke zinc (Figure 3b, left image) does not produce bright “hot spots”; this indicates that there is no significant physisorption of imaging agent **2**. Thus, the green “hot spots” that are present in Figure 3a (reaction with **1**) are assigned as oxidative addition intermediate, **6**. A small number of fluorescent “hotspots” are visible on the surface of the lithium-reduced Rieke zinc surface (Figure 3b, right image). The intensity of the green “hot spots” on the surface of sodium-reduced Rieke zinc with **1** (Figure 3a, right image) is similar to that of the control reaction (Figure

3b, right image) and is thus assigned to a small amount of physisorption.

The conclusion of the experiments with **1** and **2** is that surface organozinc intermediate **6** builds up and does not solubilize readily from the surface of sodium-reduced Rieke zinc. From the microscopy experiments alone, either of two conclusions remains plausible for the lithium-reduced Rieke zinc: (1) Lithium-reduced Rieke zinc is insufficiently reactive to produce intermediate **6**, or (2) Intermediate **6** is in fact generated but it is solubilized from the surface prior to imaging, perhaps caused by the small amount of residual LiCl in the sample. Differentiation between these two mechanistic options will be addressed in the ¹H NMR Spectroscopy for Kinetics section (wherein these latter data are consistent with conclusion 2).

The second set of experiments, in Figure 3c, compares sodium-reduced Rieke zinc (left) and lithium-reduced Rieke zinc (right), both reacting with alkyl iodide oxidative-addition partner **3**. The image of sodium-reduced Rieke zinc (Figure 3c, left image) shows bright green “hot spots.” In contrast, lithium-reduced Rieke zinc (Figure 3c, right image) only has a small number of “hot spots.”

To analyze the physical origin of the fluorescent signal in these images, control imaging agent experiments with **4** were performed under otherwise identical conditions with sodium-reduced Rieke zinc (Figure 3d, left image) and lithium-reduced Rieke zinc (Figure 3d, right image). The sodium-reduced Rieke zinc control experiment (Figure 3d, left image) displays no fluorescent “hot spots.” This indicates no significant physisorption of the imaging agent core or tether on the surface of sodium-reduced Rieke zinc. Thus, the bright “hot spots” in Figure 3c (left image) cannot be attributed to physisorption, and the signals were identified as arising from organozinc intermediate **7**. The image of the lithium-reduced Rieke zinc control experiment (Figure 3d, right image) displays a small quantity of fluorescent “hot spots,” because there is a similar amount in the right image of Figure 3c; this small amount of “hot spots” was assigned as physisorption.

Thus, it was concluded that bright spots are chemically specific to the carbon–iodide bond, and they were assigned as oxidative-addition intermediate **7**. The presence of the oxidative-addition intermediate **7** indicates that the sodium-reduced Rieke zinc is sufficiently reactive for oxidative addition with alkyl iodide **3**; however, the resulting organozinc surface intermediate is persistent on the surface of the Rieke zinc.

As with the aryl iodide experiments, the absence of observable oxidative-addition intermediate **7** on lithium-reduced Rieke zinc could be described by the two scenarios, **1** or **2**, by microscopy alone.

The last set of images, Figure 3e, show the reaction of sodium-reduced Rieke zinc (left) or lithium-reduced Rieke zinc (right) with alkyl chloride **5** at 60 °C. Carbon–chloride bonds often require elevated temperatures for insertion reactions with zinc when compared to analogous carbon–iodide bonds. For this reason, direct insertion of an alkyl chloride to zinc usually requires extreme conditions, for example, the use of a Zn(CN)₂ precursor (Figure 1b), that may produce deadly HCN during the quenching process,¹⁰ or with a higher boiling and polar aprotic solvent (e.g., DMA) coupled with elevated temperatures (e.g., 80 °C).³⁷

Yet, Figure 3e shows hotspots consistent with formation of the oxidative addition intermediate on the surface of sodium-reduced Rieke zinc. Perhaps surprisingly, this observation

indicates that the oxidative addition step of alkyl chlorides may not generally be rate-limiting, because this step of the overall process is apparently accessible under mild(er) conditions: in THF solvent at 60 °C and from Rieke zinc produced from ZnCl_2 reduction.

Figure 3f shows the corresponding control experiments, with sodium-reduced Rieke zinc (left) or lithium-reduced Rieke zinc (right) and imaging agent 4 at 60 °C. In Figure 3e (left image), very bright hot spots cover the surface of sodium-reduced Rieke zinc. In comparison, the control image in Figure 3f (left image) is darker than the left image in Figure 3e. Although bright spots occur in both, there is a significant difference in both the quantity and the brightness between Figure 3e (left image) and control Figure 3f (left image). The excess of green “hot spots” are attributed to oxidative-addition intermediate, 8.

When comparing the reaction of lithium-reduced Rieke zinc and 5 (Figure 3e, right image), with its control reaction (Figure 3f, right image), both images have similar quantities and intensities of green “hot spots.” Because these images are similar, the fluorescent bright green “hot spots” in both are attributed to physisorption and not oxidative addition intermediate 8. As with the aryl iodide and alkyl iodide experiments, the absence of observable oxidative addition intermediate 8 on lithium-reduced Rieke zinc could be described by the two scenarios, 1 or 2, by microscopy alone.

Corroborating evidence for the assignment of 7 as the persistent, relatively insoluble oxidative addition intermediate on the surface of sodium-reduced Rieke zinc is shown in Figure 4. Addition of a solution of LiCl to this sample in THF

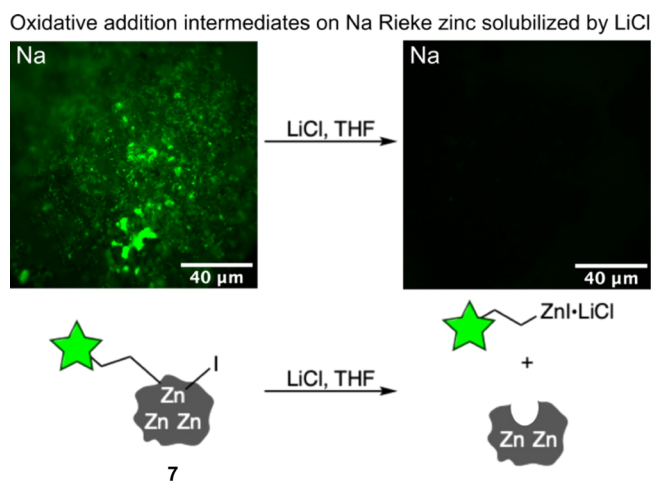


Figure 4. Fluorescence microscopy images of before and after the addition of LiCl to sodium-reduced Rieke zinc and the corresponding chemical equation.

resulted in immediate solubilization of this fluorescent material from the zinc surface, consistent with the established behavior of oxidative addition intermediates on the surface of commercial zinc metal powder.^{29–31} This solubilization step is imaged through the stark contrast between the images of the Rieke zinc sample before addition of LiCl (bright fluorescent spots present) and after addition of LiCl (fluorescent spots absent). These images show single zinc particles that are representative of the sample-wide particles and behavior throughout the sample, as seen through a microscopy sample survey.

Taken as a whole, fluorescence microscopy data of separated Rieke zinc solids show that sodium-reduced Rieke zinc solids are sufficiently reactive to accomplish the key oxidative addition/direct insertion step with aryl iodide and alkyl iodides at ambient temperature, and with alkyl chlorides at 60 °C. Yet, in all cases examined, the resulting oxidative addition intermediates are relatively insoluble and are sufficiently persistent on the surface of the sodium-reduced Rieke zinc to withstand rinsing with THF. Thus, it is plausible that the low/slow solubility of intermediates from the surface of sodium-reduced Rieke zinc (and any NaCl derived zincates formed) is a significant contributor to the lower reactivity of Rieke zinc produced this way. To our knowledge, this hypothesis has not been suggested previously in the literature.

When exogenous soluble LiCl is added to the built-up surface intermediates from sodium-reduced Rieke zinc, however, these intermediates immediately solubilize (Figure 4). This observation suggests that small amounts of residual LiCl from the preparation of lithium-reduced Rieke zinc could be solubilizing oxidative addition intermediates before they can be imaged on the surfaces of lithium-reduced Rieke zinc. Thus, residual solution-phase LiCl already present as a byproduct of the synthesis of lithium-reduced Rieke zinc plausibly accelerates the solubilization of the organozinc from the surface of the zinc^{29–31} and this faster solubilization could result in overall faster reaction rates observable on the macroscale. To test this hypothesis further, we next turned to ¹H NMR spectroscopy studies of bench-scale reactions.

NMR Spectroscopy to Characterize the Organozinc Product Structure from Different Preparations of Rieke Zinc. Although we intended to immediately study kinetics on the bench scale by ¹H NMR spectroscopy, our initial test experiments identified differences in organozinc product structure depending on the Rieke zinc preparation method as well as the presence or absence of the supernatant. While fluorescence microscopy allowed the observation of the oxidative addition intermediate, it provided no information on the solution organozinc species formed after solubilization. To complement the fluorescence microscopy experiments, ¹H NMR spectroscopy experiments were thus used to determine the species of organozinc that resulted from each form of Rieke zinc. These experiments were performed by injecting a THF-*d*₈ stock solution of 9 or 10 into Rieke zinc from various sources at ambient temperature for 60 min. Experiments were performed in triplicate.

In previous studies, the organozinc species formed depended upon whether or not LiCl was present in the reaction mixture to establish a Schlenk equilibrium (Figure 5a); the organozinc iodide 11 displays α protons at 0.51 ppm, whereas the alternative diorganozinc (13) displayed α protons shifted upfield to 0.39 ppm with a distinctive AA'XX' secondary coupling pattern, wherein the chemical shift ranges are determined by the value of halide I (Figure 5b; X = I).^{28,29} The assignment of 13 by its distinctive ¹H NMR spectroscopy signals as the diorganozinc complex is consistent with previous studies in our laboratory.²⁹ A set of DOSY NMR spectroscopy experiments were further consistent with the assignment of 13 as the diorganozinc complex: A similar (diffusion coefficient) dominant solution species was formed either in the presence of LiI or LiCl for 13, as synthesized through an alternative synthesis from commercial zinc powder and (2-iodoethyl)-benzene with LiX salt. This observation supports that the halide was not coordinated to the dominant solution species,

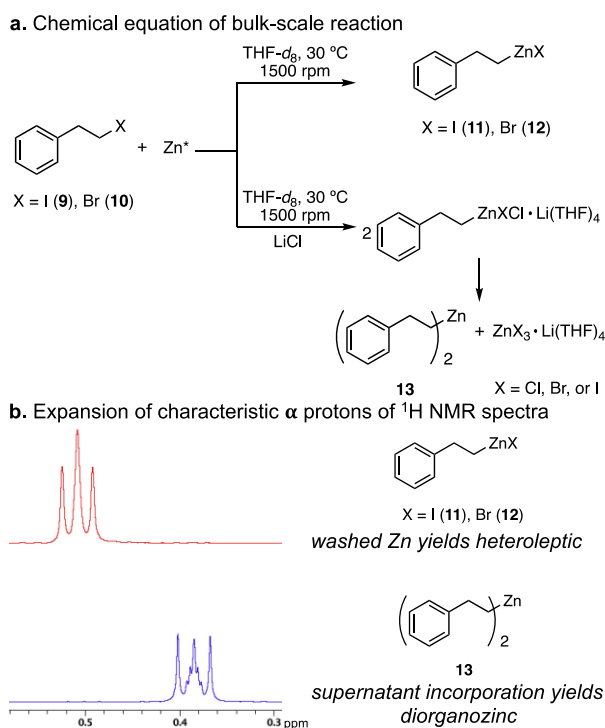


Figure 5. (a) Chemical equation of bulk-scale organozinc synthesis in the presence and absence of lithium chloride; (b) expansion of the distinctive α proton region of the synthesis of an organozinc with alkyl iodide with Li Rieke zinc when the zinc is washed to remove the supernatant (top, heteroleptic organozinc) and when the supernatant from the synthesis of the Rieke zinc is not removed via washing and remains present in the following reaction (bottom, diorganozinc), showing formation of two different species.

because its diffusional behavior was independent of halide. Therefore, these data further support the assignment of the composition of the dominant organozinc solution species as diorganozinc **13**. (see SI, Section 10.4, for DOSY experiments.)

In the current study, the α proton spectral region varied depending on whether or not the residual salts in the supernatant from Rieke zinc synthesis were included in the reaction. When residual chemical species from the supernatant were included in the reaction from the lithium-reduced Rieke zinc synthesis, diorganozinc **13** forms, as detected by its distinctive ^1H NMR spectrum (Figure 6). The Schlenk equilibrium positions of monoorganozinc and diorganozinc in THF solvent have been previously characterized by Organ in the presence of LiBr and found to favor diorganozinc species, similar to this assignment.¹¹

Supernatant determines organozinc species

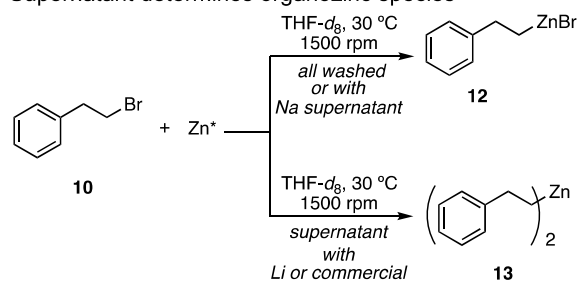


Figure 6. Organozinc synthesis with and without the supernatant.

However, when the supernatant is removed by washing prior to addition of the organohalide, such that only the solids from the lithium-reduced Rieke zinc were employed in the reaction **11** (or **12**) is detected by its distinctive triplet α protons at 0.51 ppm (Figure 5b).²⁹ A similar outcome occurred when sodium-reduced Rieke zinc was used, with or without its supernatant.

NMR Spectroscopy for Kinetic Studies of Reactivity Rate Differences between Different Preparations of Rieke Zinc. As mentioned in the introduction, Rieke's reports of reactivity differences were largely qualitative. To our knowledge, there is no prior report, qualitative or quantitative, comparing rates of the same direct insertion reaction in the presence and absence of the residual supernatant from one preparation of Rieke zinc. One reason for the absence of this quantitative data may be the prior prevailing thought that the solids were the dominant source of the reactivity differences from different preparations. To fill this gap in knowledge, we performed these kinetics experiments.

Kinetic data were collected on the formation of organozinc using an alkyl halide (iodide **9** or bromide **10**) and various forms of Rieke zinc (sodium-reduced, lithium-reduced, or commercially purchased, each with and without the supernatant). Purchased Rieke zinc (Rieke Metals) was synthesized commercially by reduction of zinc(II) chloride with lithium.^{17,19,24} These data were used to evaluate whether or not the supernatant and the byproduct salts it contained from the synthesis of Rieke zinc affected bench-scale reaction rates in the subsequent synthesis of organozinc reagents by direct insertion of organohalides. Each form of Rieke zinc reaction mixture was measured by ^1H NMR spectroscopy in triplicate at 15, 30, and 60 min using mesitylene as an internal standard.

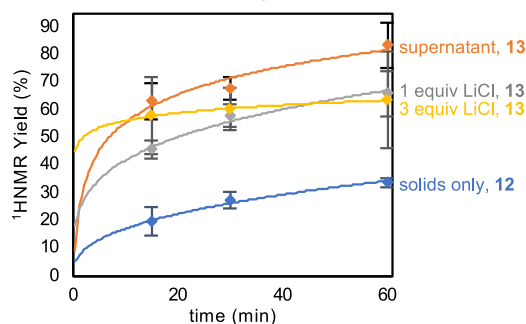
Because reactions using alkyl iodide **9** proceeded too rapidly for facile monitoring by ^1H NMR spectroscopy (see the SI for details), slower alkyl bromide **10** was used as the substrate. To enable detection by ^1H NMR spectroscopy, reaction concentrations were significantly higher in kinetics experiments than in prior sensitive microscopy experiments, accounting for these fast reactions on the bench scale with alkyl iodides.

Four different ^1H NMR spectroscopy kinetics experiments were performed with commercially purchased Rieke zinc and alkyl bromide **10** (Figure 7a; fits determined empirically): (1) The orange data in Figure 7a show the formation of **13** from commercial Rieke zinc and alkyl bromide **10** when the supernatant is included in the reaction (orange line shows power fit). (2 and 3) The yellow and gray data show the reaction of washed commercial Rieke zinc and **10** with 3 equiv or 1 equiv, respectively, of exogenous LiCl added (lines show power fit). (4) The blue data in Figure 7a show the reaction of the commercial Rieke zinc solids (without any added LiCl or supernatant (blue line shows power fit). The error bars show the range from triplicate runs and reflect the variation inherent in this heterogeneous reaction mixture (see the SI for details).

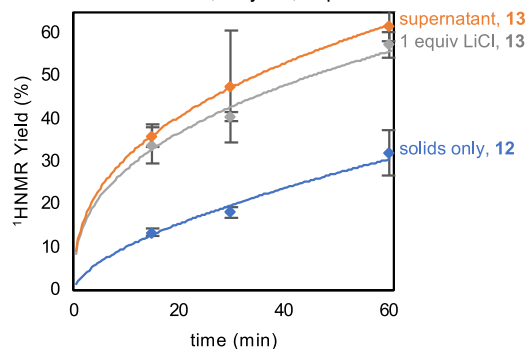
These kinetic profiles show the dramatic effect of solution-phase LiCl. When either solid LiCl (yellow and gray line, Figure 7a) or the previously separated supernatant containing LiCl (orange line, Figure 7a) was added to the reaction, and the reaction rates were accelerated compared to the reaction rate of the solids alone. In the case of the solids alone, the reaction was markedly slow and low yielding and resulted in organozinc halide **12**. This lower solubility of NaCl and the presumed similar lower solubility of the resulting sodium salts of any NaCl derived zincates thus result in reduced formation

Kinetics of bench-scale synthesis show effects of LiCl

a. Commercial Rieke zinc, alkyl Br; supernatant contains LiCl



b. Li-reduced Rieke zinc, alkyl Br; supernatant contains LiCl



c. Na-reduced Rieke zinc, alkyl Br; supernatant contains NaCl

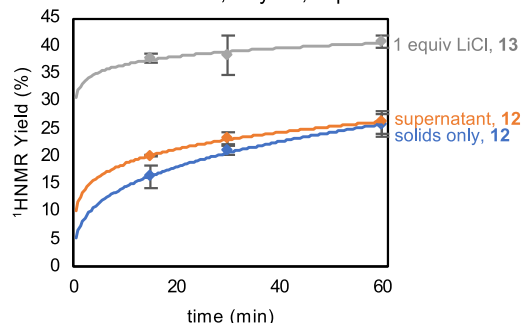


Figure 7. (a–c) Kinetics data for alkyl bromide **10** and Rieke zinc from various sources (with and without supernatant, and/or with added LiCl). In each graph, the gray line shows the organozinc formation with the addition of 1 equiv LiCl as an additive, the orange line shows organozinc formation with the residual salts from the Rieke zinc reaction mixture added in as an additive, the blue line shows organozinc formation with washed Rieke zinc with no additives, and the yellow line shows organozinc formation with three equiv of LiCl added into the reaction mixture as an additive.

of **13** from sodium-reduced Rieke zinc. In contrast, the faster reactions in the presence of LiCl (either added as a solid or in solution via the supernatant) produced diorganozinc **13**.

Figure 7b features the kinetic profiles of the reactions between house-made lithium-reduced Rieke zinc and alkyl bromide **10** with various additives. Again, soluble LiCl caused a striking rate acceleration. The orange data show the reaction between lithium-reduced Rieke zinc in the presence of its supernatant and **10**. In comparison, the rate and yield were far decreased when only the solids of lithium-reduced Rieke zinc were employed (blue data, Figure 7b).

This rapid rate could be restored in the absence of supernatants by simple addition of exogenous soluble LiCl. When 1 equiv of LiCl was added to the reaction of the solids of

lithium-reduced Rieke zinc and **10** (gray line, Figure 7b), the rate of reaction was increased to a similar rate of the reaction with the addition of the supernatant (orange data, Figure 7b). In all cases, lines show power fits. The reactions of house-made sodium-reduced Rieke zinc and **10** are different from the previous two preparations of Rieke zinc because the supernatant contains the NaCl byproduct from reduction (instead of LiCl). In these cases, the presence of the supernatant failed to cause a difference in rates—this composition distinction pinpoints LiCl as the specific actor for the increase in reaction rates (orange and blue data, Figure 7c).

Regarding the structure of the organozinc product, even when the NaCl containing the supernatant was added, organozinc halide **12** was formed, consistent with the absence of the LiCl-specific-driven Schlenk equilibrium.²⁹ Notably, exogenous LiCl, although never “naturally” present in the original reaction, caused a dramatic rate acceleration and yield enhancement for sodium-reduced Rieke zinc. The conclusions of the ¹H NMR spectroscopy kinetics experiments are that residual lithium chloride in the supernatant substantially increases the rate of formation of organozinc reagents. This lithium chloride is present “naturally” during synthesis as a byproduct of lithium reduction of zinc chloride. Similar rate accelerations, however, are achievable by addition of LiCl to Rieke zinc prepared by an alternative preparation that lacks byproduct LiCl.

Solid Rieke Zinc Surface Characterization. The identification of rate enhancement by residual salts in the supernatant does not, on its own, preclude salts in the solids also contributing to rate enhancement. To gauge the plausibility of this additional contributing factor, the compositions of the solids from different preparations of Rieke zinc were next examined. These composition studies aimed to determine if sufficient/substantial quantities of salts were also present in the solids, when compared to the quantity of salts in the solutions.

SEM analysis identified different surface morphologies of Rieke zinc from different preparations (Figure 8a–c). The images show textured surfaces of lithium-reduced and commercial Rieke zinc (indicative of a high surface area), and of sodium-reduced Rieke zinc (Figure 8a–c). This increased surface area of lithium-reduced Rieke could be responsible for the increased physisorption of those solids as detected by fluorescence microscopy (Figure 3b,d,f).

SEM, EDS, and XPS experiments were performed for elemental analysis and/or surface-morphology characterization of the surface of each form of zinc solids (separated from their respective supernatants). EDS data were consistent with the bulk of all three samples containing predominantly zinc (Figure 8a–c) but proved unhelpful for further elemental characterization. Therefore, XPS was performed as a complementary technique. All samples displayed two XPS peaks for zinc, consistent with incomplete reduction, and peaks for chloride and carbon (Figure 8a–c). Additionally, sodium was detected in the sample reduced with sodium (Figure 8a). No lithium was detected, however, in the lithium-reduced Rieke zinc or commercial Rieke zinc (Figure 8).

The conclusion from these surface composition studies is that sodium is present on the surface of sodium-reduced Rieke zinc, but there is no detectable lithium on the surface of lithium-reduced Rieke zinc. This observation of no or minimal lithium is inconsistent with the idea that Rieke zinc solids are the predominant source of lithium chloride that leads to rate

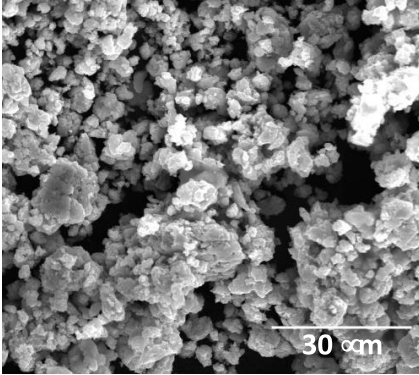
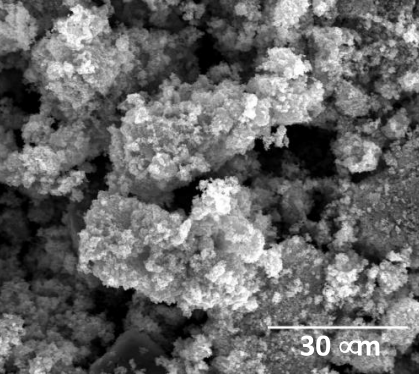
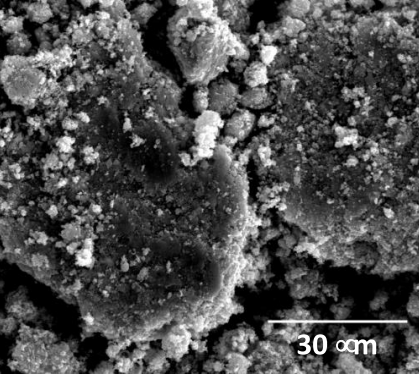
a. sodium-reduced Rieke zinc	b. lithium-reduced Rieke zinc	c. commercially purchased Rieke zinc
EDS: 55% Zn XPS: Zn, Cl, C, Na ICPMS: 1:7 (Na:Zn) ICPMS*: 1:3 (Na:Zn)	EDS: 49% Zn XPS: Zn, Cl, C ICPMS: 1:340 (Li:Zn) ICPMS*: 1:10 (Li:Zn)	EDS: 79% Zn XPS: Zn, Cl, C ICPMS: 1:38 (Li:Zn) ICPMS*: 1:10 (Li:Zn)
		

Figure 8. (a) EDS, X-ray photoelectron spectroscopy (XPS), ICP-MS, ICP-MS* (*denotes supernatant incorporation), and SEM of sodium-reduced Rieke zinc. (b) EDS, XPS, ICP-MS, ICP-MS* (* denotes with supernatant incorporation), and SEM of lithium-reduced Rieke zinc. (c) EDS, XPS, ICP-MS, ICP-MS* (* denotes with supernatant samples), and SEM of commercially produced Rieke zinc. All SEM images were taken with 1600 $\times$ magnification and a 6.22 nA current.

enhancement of lithium-reduced Rieke zinc. It remained possible, however, that surface compositions were not representative of bulk solid samples. Therefore, elemental analyses of bulk samples were next performed.

Solid Bulk Sample Characterization. ICP-MS analysis was performed on two series of solids for comparison: (1) Zinc solids alone, consisting of isolated solids of Rieke zinc from which the supernatant had been removed and (2) zinc solids with supernatant residue, consisting of solids of Rieke zinc to which an equivalent amount of the supernatant solution from the corresponding Rieke zinc preparation reaction (which had been previously separated) was added back in. In this “with supernatant-series,” the THF solvent was removed via reduced pressure under inert atmosphere prior to ICP-MS analysis of the samples, leaving behind the nonvolatile components in the supernatant for analysis. Samples were prepared under an inert atmosphere, removed from the inert atmosphere immediately before digestion, and digested with hydrochloric acid and nitric acid prior to analysis.

This analysis showed minimal lithium present in the solid-only sample of the lithium-reduced Rieke zinc as seen by the low parts ratio of lithium-to-zinc (Li:Zn, 1:340; Figure 8b). In contrast, when the supernatant was added back in, substantial lithium was present as seen by the increase in ratio of lithium-to-zinc (Li:Zn, 1:10). This measurement established unequivocally that most residual lithium was present in the supernatant and not in the solid Rieke zinc.

Commercial Rieke zinc showed a similar analysis pattern. Zinc solids alone contained a 1:38 Li:Zn ratio, whereas when the supernatant was added back in, lithium composition rose to a 1:10, Li:Zn ratio (Figure 8c).

More NaCl was present in the sample with the supernatant added back in (1:3, Na:Zn), than in the isolated zinc solids alone (1:7, Na:Zn); however, the similarity in these two ratios apparently arises from the relative insolubility of NaCl in THF, leading to more residual salt composition in the sodium-reduced solids than in the comparable lithium-reduced cases.

Together, surface and bulk analysis is consistent with the idea that the high solubility of LiCl in THF results in minimal LiCl becoming imbedded in the Rieke zinc solids during preparation. In contrast, the low solubility of NaCl in THF results in substantial NaCl becoming imbedded in Rieke zinc solids during preparation reactions. It is this soluble byproduct LiCl in the supernatant that causes the otherwise persistent surface organometallic intermediate to solubilize, as seen by fluorescence microscopy (Figures 3 and 4). The rapid solubilization by LiCl of these intermediates is then the mechanistic cause of the bench-scale rate increases measured by ^1H NMR spectroscopy (Figure 7).^{29,32}

CONCLUSIONS

In stark contrast to previous understanding, these data show that notable differences in reactivity from different preparations of Rieke zinc arise from residual salts in the supernatant of the most-often used preparations of Rieke zinc. Through fluorescence microscopy with single-Rieke-zinc-particle resolution, the mechanistic origin of these differences in reactivity has been identified as differences in persistence of the surface organozinc intermediates (Figures 3 and 4). This supernatant dictates not only overall reaction kinetics but also the composition and structure of the resulting organozinc species. A summary of these findings is represented in Figure 9.

This understanding enables the experimentalist to “dial in” a product selection for the mono- or diorganozinc complex (with concurrent Schlenk equilibrium) through simple selection or addition of soluble LiCl salt. Each complex is anticipated to display unique reactivity and chemoselectivity (e.g., in palladium-catalyzed cross-coupling reactions).^{35,36,38–40} Interestingly, some preparation methods by Rieke describe removal of the supernatant for swap of the organic solvent from THF to an alternative.^{4,9,17} Users of this methodology have presumably not known they were changing the structure of the resulting reagent by choice of preparation method. These results go against prevailing hypotheses since

Organozinc species and rate depends on presence of supernatant

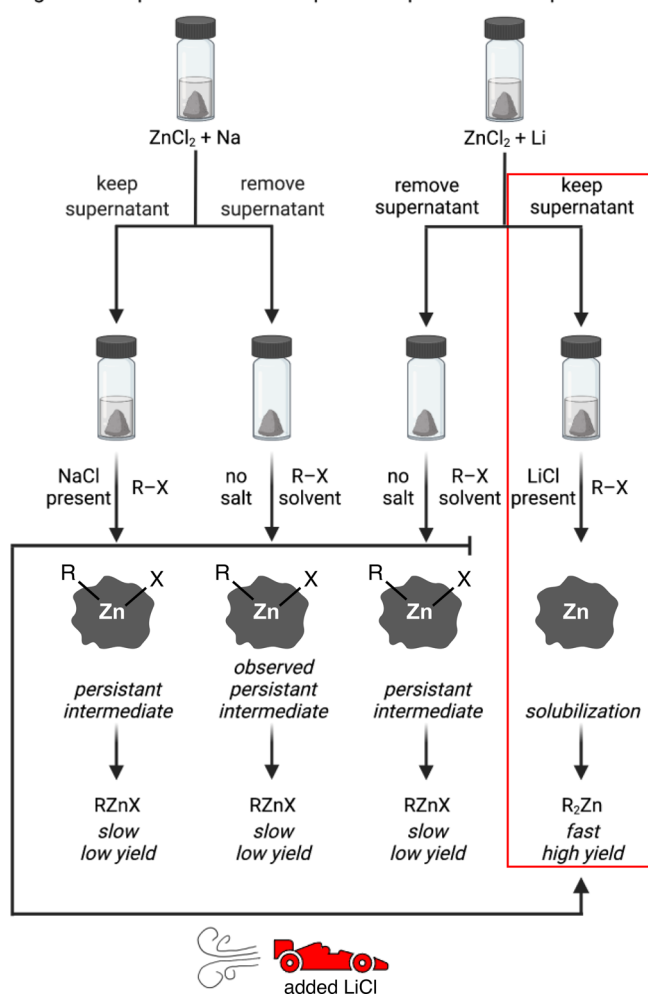


Figure 9. Summary of the findings of this work.

the 1970s for synthetic preparations that continue to see high use today.^{21,25,26,41} Reactivity conclusions may extend to other sources of finely divided zinc (i.e., “nanozinc”), as employed in the presence of salts recently in modern organic synthesis.⁴² These experiments thus provide a new understanding that reactivity improvements for finely divided metals are accessible through manipulation of the supernatant composition during metal preparation—which is a much easier and more accessible manipulation than altering the metal lattice structure and composition.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.2c02471>.

Detailed experimental procedures, replicate fluorescence microscopy data, replicate kinetics experiments, additional SEM images, EDS, XPS, and ICPMS data (PDF)

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Notes

The authors declare no competing financial interest.

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